Caught & Shared: Conscious Control in Recommender Systems

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CCS Concepts

• **Information systems** → **Recommender systems**; *Personalization*; • **Human-centered computing** → *Collaborative and social computing*.

Keywords

recommender systems, personalization, mentor-based recommendation, user-driven recommendation, conscious recommendations, human-in-the-loop

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1 ABSTRACT

The core insight behind **Caught & Shared** is disarmingly simple: sometimes the best recommendations aren't the ones that look "most similar to me", but the ones that come from someone whose taste I genuinely admire.

Recent industry developments suggest growing interest in more transparent and user-influenced recommendation experiences.

Caught & Shared takes this trend further by introducing conscious, user-driven recommendation control. Instead of passively accepting what the algorithm thinks is best, the user can deliberately borrow someone else's taste. This approach has the potential to improve discovery, exploration, and user engagement while introducing a new form of conscious participation in recommendation systems. It transforms recommendations into a more transparent and human one.

In the broader vision of *Conscious Recommendations* (discussed in Section 6), Caught & Shared becomes one of the foundational features — a practical first step toward giving users real agency over their digital experience.

Supplementary Materials: https://rjeyskey.github.io/Caught-and-Shared

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2 INTRODUCTION

Today's recommender systems have reached remarkable levels of sophistication. They can build detailed profiles of our tastes, preferences, and behaviors. Yet users rarely have direct ways to influence recommendation logic beyond their behavioral signals.

One of the most natural forms of communication between people is sharing taste. When we see a friend's exceptionally good feed, a strong desire arises: "I want something like this for myself." Not merely similar to my existing preferences, but influenced by someone whose taste I trusted. This feeling is familiar to many. Recently, while looking at my classmate's Avito (a large Russian classifieds and marketplace platform) feed, I thought: "Damn, this is so much better than mine." His recommendations contained something deeply genuine — they reflected how he sees the world, what excites him, and what matters to him. At that moment, I didn't want another algorithmic guess. I wanted *his* lens.



Figure 1: Conceptual example of trust-driven recommendation transfer, where users intentionally adopt recommendation influences from people whose taste they value.

Current recommender systems largely assume that preference discovery should happen implicitly. But what if recommendations became a space where users could take an active, conscious part in shaping their own experience?

Sometimes a user doesn't fully understand their own preferences. Even when they have a vague sense of what they

want, they often don't know *how* to configure the recommendation mechanism to achieve it. However, they can clearly recognize whose recommendations resonate with them. In such cases, the most honest and effective solution is simple: let them point to a specific person and say — "Make it more like this."

Caught & Shared explores exactly this idea. It transforms recommendations from a purely algorithmic black box into a tool of conscious self-expression and human connection. Because sometimes the most powerful way to understand another person — or to let others understand you — is not through words, but through what you choose to watch, read, and buy.

This is more than a new feature. It is the beginning of a new paradigm — **Conscious Recommendations**, where the user is no longer just a passive consumer of algorithms, but an active curator of their own information world.

3 RELATED WORK

Modern recommender systems have evolved from collaborative filtering toward graph-based, sequential, and foundation-model architectures. However, the dominant paradigm remains *implicit personalization*, where the system attempts to infer user preferences without direct user participation in the recommendation process.

Graph Neural Networks (GNNs) have become a cornerstone of modern recommendation. Works such as Neural Graph Collaborative Filtering (NGCF) [3], PinSage [1], and TwHIN [2] demonstrated the power of modeling high-order user-item relationships. Social recommendation literature, notably GraphRec [4] and DANSER, further explored the integration of social connections to enhance recommendation quality. Nevertheless, these approaches typically treat social links as static and passively learned rather than allowing users to *explicitly* designate trusted sources of taste.

In parallel, contrastive learning has emerged as a powerful paradigm in self-supervised recommendation. Frameworks such as SGL [5] and various CL4Rec-style models apply contrastive objectives to learn more robust user and item representations. While effective at improving embedding quality, these methods still operate within the traditional similarity-based paradigm and do not provide mechanisms for users to intentionally incorporate specific human perspectives.

Early research on mentor-based recommendations dates back to Zhang et al. (2011), who proposed identifying influential "leaders" or "mentors" to alleviate the cold-start problem. More recent works have explored knowledge transfer across users and domains. However, the majority of these studies focus on *automatic* mentor detection rather than giving users conscious, explicit control over the process. Brun et al. [12] showed that a small set of mentors (6) suffices to explain the preferences of a large user base, and that mentors can be selected from behavior graphs without explicit ratings.

Recent platform developments also indicate growing interest in user participation. TikTok has experimented with collaborative recommendation experiences through Shared Feed concepts, while Instagram introduced mechanisms allowing users to inspect and influence recommendation signals. Yandex Music's *Feed for Two* feature goes further by letting two close people actively shape a shared recommendation space, rooted in the bond between them — an industry example of user-driven, relationship-centered taste control. Although these systems differ substantially from Caught & Shared, they reflect a broader industry trend toward greater transparency and user involvement in recommendation processes.

The concept of **Conscious Recommendations**—where users actively participate in shaping and directing their recommendation environments—remains relatively underexplored in academic literature. While modern recommender systems provide increasingly sophisticated personalization mechanisms, user influence is still predominantly indirect and mediated through interaction signals. Our work explores an alternative perspective through **Caught & Shared**, a framework that enables explicit, controllable, and multi-faceted mentor integration through six distinct technical approaches: Heterogeneous Graph, Virtual Proxy Items, Dual Graph, MetaPath-based Propagation, Contrastive Alignment, and Embedding Blending.

This work explores the idea of user-driven mentor mechanisms across multiple recommendation architectures and introduces the concepts of category-aware mentor assignment and adjustable influence levels. Rather than presenting a single implementation, we aim to outline a broader design space in which users can consciously participate in shaping their recommendation environments. We hope that these ideas contribute to the ongoing discussion around more transparent, controllable, and human-centered recommender systems, which we refer to as **Conscious Recommender Systems**.

4 CORE IMPLEMENTATION TAXONOMY

To realize the **Caught & Shared** concept, we propose three universal strategies for injecting mentor influence into recommender systems. These strategies are designed to be adaptable to different backbone architectures. Figure 2 illustrates four concrete implementations within the graph-based family.

4.1 Graph-based (Structural) Approaches

This family integrates mentor information directly into the graph representation, enabling natural propagation of taste through network structure.

- **Heterogeneous Graph**: The most straightforward realization. We augment the original user-item bipartite graph with explicit directed *user* → *mentor* edges.

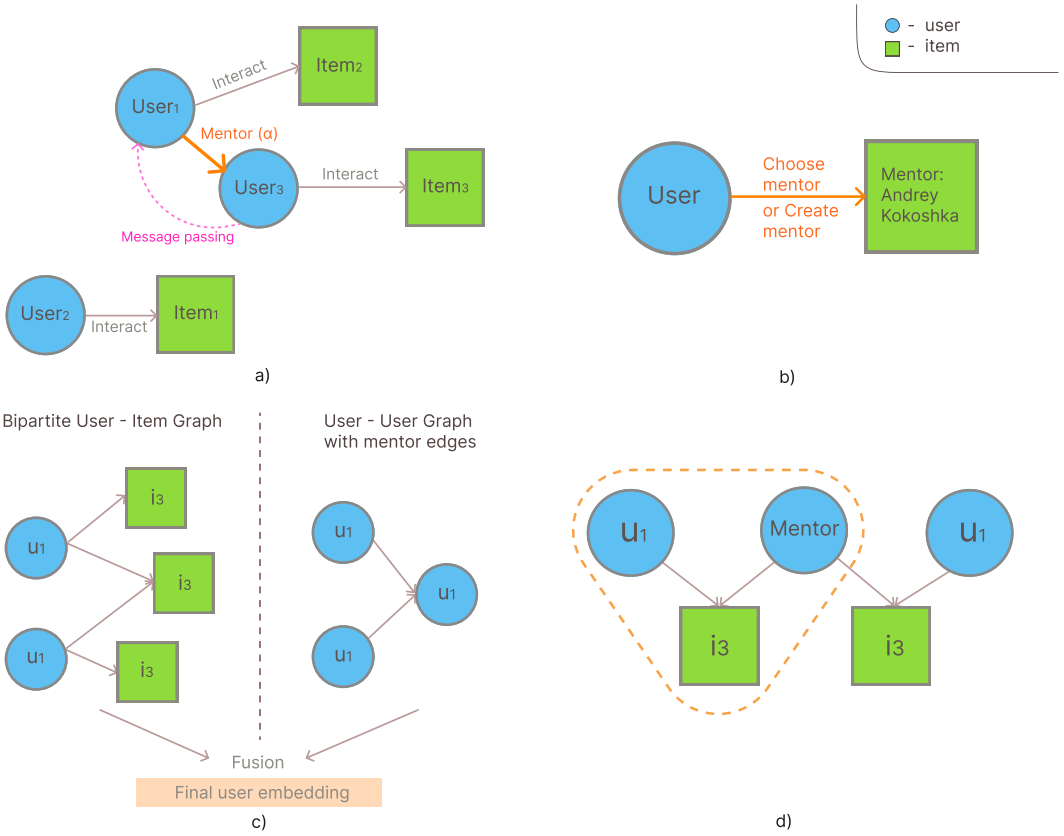


Figure 2: Graph-based approaches for mentor influence: (a) Heterogeneous Graph, (b) Virtual Proxy Items, (c) Dual Graph, (d) MetaPath-based.

This transforms the graph into a heterogeneous multi-relational structure with two types of relations: *interact* (user–item) and *mentor* (user–user). Message passing mechanisms (e.g., in R-GCN or HGT) then naturally propagate

preferences from the mentor to the target user.

- **Virtual Proxy Items (Node Transformation)**: To preserve the bipartite structure, each mentor is represented as a special virtual item. Selecting a mentor is modeled as an interaction with this proxy item. This approach allows the use of standard bipartite GNNs such as LightGCN without architectural modifications.
- **Dual Graph**: We maintain two separate graphs — a classical bipartite user-item graph and a social user-mentor graph. User embeddings are learned independently on each graph and then fused (via concatenation, attention, or weighted sum), providing modular and flexible integration of mentor signals.
- **MetaPath-based**: Without adding new edges, we define semantically meaningful metapaths (e.g., User → Mentor → Item or User → Item ← Mentor → Item). The model aggregates information along these paths, allowing mentor influence while strictly preserving the original bipartite structure. This approach offers high interpretability and compatibility with models such as HAN and MAGNN.

4.2 Contrastive Alignment

This approach operates at the embedding level without modifying the graph. During training, we introduce a contrastive objective that pulls the user's embedding closer to their chosen mentor's embedding in latent space, while pushing it away from random users (using losses such as InfoNCE). It is highly flexible and can be applied on top of any neural architecture, including Two-Tower models and GNNs.

4.3 Embedding Blending (Composite)

The most lightweight and user-controllable method. At inference time, we directly blend the user's embedding with the mentor's embedding using a controllable coefficient α:

$$\mathbf{u}_{\text{final}} = (1 - \alpha) \cdot \mathbf{u} + \alpha \cdot \mathbf{m} \tag{1}$$

This technique provides maximum transparency and works as a plug-in layer on top of virtually any existing recommender model.

4.3.1 Why we prefer it, and what a system should do to adopt it Embedding Blending is deliberately the approach we would prototype first, because it is the only one of the three that is:

- **Non-invasive** — it sits on top of an existing engine without retraining it or touching its internal representation ("we don't ask you to change your core RecSys").
- **Controllable** — α is a real, user-visible dial (Light 20%, Balanced 45%, Deep 70%).
- **Transparent** — the outcome is explainable to a user: "your feed is 40% your friend's taste."
- **Reversible** — $\alpha = 0$ reproduces the exact original feed, so rollback is guaranteed by construction (never a silent takeover).

This directly implements the user-agency principles at the heart of Caught & Shared, and it is the easiest of the six approaches to integrate with an existing production pipeline: it needs (1) a way to obtain the user's and the mentor's embeddings at serving time, (2) a small per-user store of (mentor, category, α) assignments, and (3) an inference-time blend step before candidate ranking. Such a layer can be introduced gradually and evaluated with a controlled experiment; the broader safety and benefit questions then become empirical.

4.3.2 What we plan to measure (and what the literature already supports) We intend to validate Caught & Shared with a small, controlled user study in which participants choose *their own* friends as mentors, tune α, and report both objective (saved-item ranking) and subjective (perceived control) outcomes. - User control over recommendation parameters measurably improves how users evaluate the results [13]. - Transparent, editable user profiles let people improve their own recommendations in a realistic setting [14]. - The optimal social-learning strategy for matter-of-taste domains depends on experience and niche-ness [16], which motivates exposing the strategy choice to the user (the α dial instead of a hidden similarity estimate). - Modeling a user as multiple interest vectors with a controllable trade-off (ComiRec) is a proven industrial direction [15].

We also stress that the cultural foundation — trusting specific people for specific domains — is a familiar human behavior (we already ask friends for music, films, and buying advice), which the above references substantiate.

All three strategies can be further extended with multi-mentor and multi-category support, leading to the broader vision of **Conscious Recommendations** (discussed in Section 6), where users build personalized "Mentor Teams" across different spheres of their life.

5 BENEFIT CONSIDERATIONS

Caught & Shared introduces a new form of user participation in recommender systems. Instead of relying exclusively on implicit interaction data, users can explicitly indicate whose recommendations, preferences, or taste they trust.

This approach may provide several benefits.

First, it increases user agency. Modern recommender systems are highly effective at inferring preferences from historical interaction patterns, yet users typically have limited direct influence over recommendation logic. Caught & Shared introduces a mechanism through which users can consciously shape part of their recommendation environment by selecting trusted sources of influence. The literature confirms that giving users explicit control over a recommendation parameter improves how they evaluate the output [13], and that inspectable, editable user models increase perceived usefulness and satisfaction [14].

Second, it enables a different form of discovery. Recommendations influenced by trusted individuals may be perceived differently from purely algorithmic suggestions. When users know that a recommendation space has been shaped, at least partially, by someone they respect or admire, they may become more willing to explore unfamiliar content, products, or ideas.

Third, recommendation profiles become a form of communication. Taste often reflects values, interests, aspirations, and identity. Sharing recommendation profiles may therefore provide a lightweight way for people to express themselves and help others discover content through their perspective.

The concept may also create opportunities for platforms. Explicit trust relationships could serve as an additional signal alongside conventional interaction-based signals, potentially improving recommendation quality, user engagement, and satisfaction. Furthermore, recommendation sharing introduces new social interactions that extend beyond conventional content sharing.

Naturally, such systems would introduce new challenges related to influence, transparency, privacy, and misuse. However, the potential benefits appear substantial enough to justify further exploration of user-driven recommendation paradigms.

6 FUTURE GROWTH TOWARD CONSCIOUS RECOMMENDER SYSTEMS

6.1 The Emerging Hype: Conscious Recommendations

One possible bright direction for the future evolution of recommender systems is what we refer to as **conscious recommendations**.

Unlike traditional personalization, where users passively receive algorithmically curated content, conscious recommendations put users in the driver's seat. They can see why certain items appear, whose taste is influencing the feed,

and — most importantly — they can actively shape their recommendation experience. This trend is driven by growing user fatigue with filter bubbles, a desire for serendipity, and the recognition that one-size-fits-all personalization fails to capture the multifaceted nature of human interests.

The **Caught & Shared** concept sits at the very core of this paradigm, offering a simple yet powerful mechanism: borrowing a trusted user's recommendation settings.

6.2 The Timur Case

A real-life example illustrates this need perfectly. My friend Timur, an advanced user of Avito, maintains three separate accounts. Under each account he gets a noticeably stronger, more refined and nuanced feed.

The reason is simple: Timur isn't interested in shallow, average recommendations. He wants to go deeper. He wants to see the *best* items in each category — the ones that give him that strong emotional response. Having separate accounts allows him to escape the noise of mediocre suggestions and focus only on higher-quality, more interesting things. For him, this approach is significantly better both in terms of time (he doesn't waste it scrolling through "common" items) and emotional satisfaction.

Modern recommendation systems already recognize that users have multiple interests. The challenge is not recognizing those interests, but optimizing them independently. A single feed often mixes fashion, technology, books, collectibles, and dozens of other domains into one recommendation space. Timur's behavior suggests that users may sometimes prefer highly specialized recommendation environments, each optimized for a distinct goal or interest area. Pinterest boards represent an interesting step in this direction by allowing users to separate contexts. However, the broader opportunity may lie in recommendation systems that can naturally support multiple recommendation identities, each with its own objectives, signals, and trusted sources of taste.

Beyond separate accounts, Yandex Music's *Feed for Two* is another example of taste shared with a close person — not a co-owned feed in the ownership sense, but a bond-based, user-driven space where the mix of two tastes becomes an expression of the relationship itself. It shows both the appeal and the limits: the user still cannot explicitly control *whose* taste dominates or undo the mix step by step — exactly the gap Caught & Shared fills with a visible α and rollback.

6.3 The Vision: Multi-Category Mentors

A natural extension of Caught & Shared is to move beyond a single borrowed feed and toward *category-specific* sources of influence. Most people do not live inside one unified "taste"—their lives are naturally split into domains: hobbies, home, work, clothing, entertainment, family, and more. What changes is not the existence of multiple interests (modern systems already model that reasonably well), but the possibility of shaping each domain *independently* and *intentionally*.

In practice, many of us already have "high-trust" people for particular areas. One friend consistently finds great music, another has exceptional taste in films, and someone else always discovers interesting items to buy. The idea that different parts of your life could be guided by different people you respect is, in a surprising way, simply *warm*. It turns recommendations from a single opaque stream into a set of relationships and deliberate choices.



Figure 3: A conceptual vision of recommendation systems where different areas of life are guided by different sources of trusted taste.

This is the direction where the Caught & Shared concept naturally leads: not to copying someone else's feed, but to a personal dashboard of taste, where each life domain can have its own mentor (or mentor team), its own signals, and its own recommendation dynamics. Rather than hoping that one global feed will satisfy every intent, users could "pick the lens" that fits the moment—and switch lenses as their goals change.

This perspective also reframes the platform–user relationship. Users are not passive. They already know meaningful things about themselves: what they care about, what they are trying to improve, and whose judgment they trust in specific contexts. Allowing that knowledge to participate in the system—explicitly and transparently—can be seen as a form of "trust teamwork": the platform contributes scale and inference, while the user contributes intent and trusted human perspective. The goal is not to show everything a user might like, but to help them consistently reach *the best* within the domains they care about.

Overall, we believe that user-driven recommendation control is a promising direction with potential value for both users and platforms. This is why we continue exploring the broader paradigm we refer to as **Conscious Recommender Systems**.

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